



# FutureProof — Equity Preservation Mortgage

Investment Opportunity Overview  
Quantitative Risk Analysis | February 2025

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## Executive Summary

FutureProof has developed an innovative financial product — the Equity Preservation Mortgage (EPM) — that allows homeowners to access their home equity while retaining full ownership and upside participation. Independent quantitative modelling confirms the product is commercially viable with appropriate structuring.

### Key Investment Highlights

1. **Large addressable market** — Australian homeowners aged 55-75 with \$2M+ property values represent a \$500B+ equity pool. Current reverse mortgage penetration is <1%.
2. **Validated product economics** — 50,000-path Monte Carlo simulation confirms deficit probability as low as 0.5% with P&I structure and buffer ETF allocation.
3. **Multiple revenue streams** — origination fees, ongoing margin, profit sharing, and insurance premium income.
4. **Institutional partnerships** — BlackRock provides fund management with 10% expected return / 10% volatility assumptions, removing investment performance risk from FutureProof.
5. **Defensible IP** — proprietary holiday mechanism, equity preservation structure, and quantitative risk engine create significant barriers to entry.
6. **Scalable model** — marginal cost per EPM decreases as the portfolio grows; insurance risk diversifies across cohorts.

# The Market Opportunity

## The Problem

- Australian homeowners over 60 hold ~\$1 trillion in residential property equity
- Most have limited income in retirement but are asset-rich
- Traditional reverse mortgages have a negative reputation (compound interest, no upside, estate erosion)
- Home downsizing has significant emotional, social, and tax implications

## The FutureProof Solution

- Homeowner borrows against equity → funds invested in managed equity portfolio
- Investment returns pay the mortgage interest + generate an annuity income
- Homeowner keeps the house AND participates in equity market upside
- Holiday mechanism protects during market downturns
- At maturity/exit: surplus (investment minus loan) returned to homeowner/estate

## Market Sizing (conservative estimates)

| Metric                   | Estimate                                   |
|--------------------------|--------------------------------------------|
| Target demographic       | Homeowners aged 55-75, property \$1.5M+    |
| Addressable households   | 200,000+                                   |
| Average EPM size         | \$1.0-1.5M                                 |
| Total addressable market | \$200-300B in loan originations            |
| At 1% penetration        | \$2-3B portfolio, ~\$40-60M annual revenue |

## How It Works — Product Overview

### Product Structure

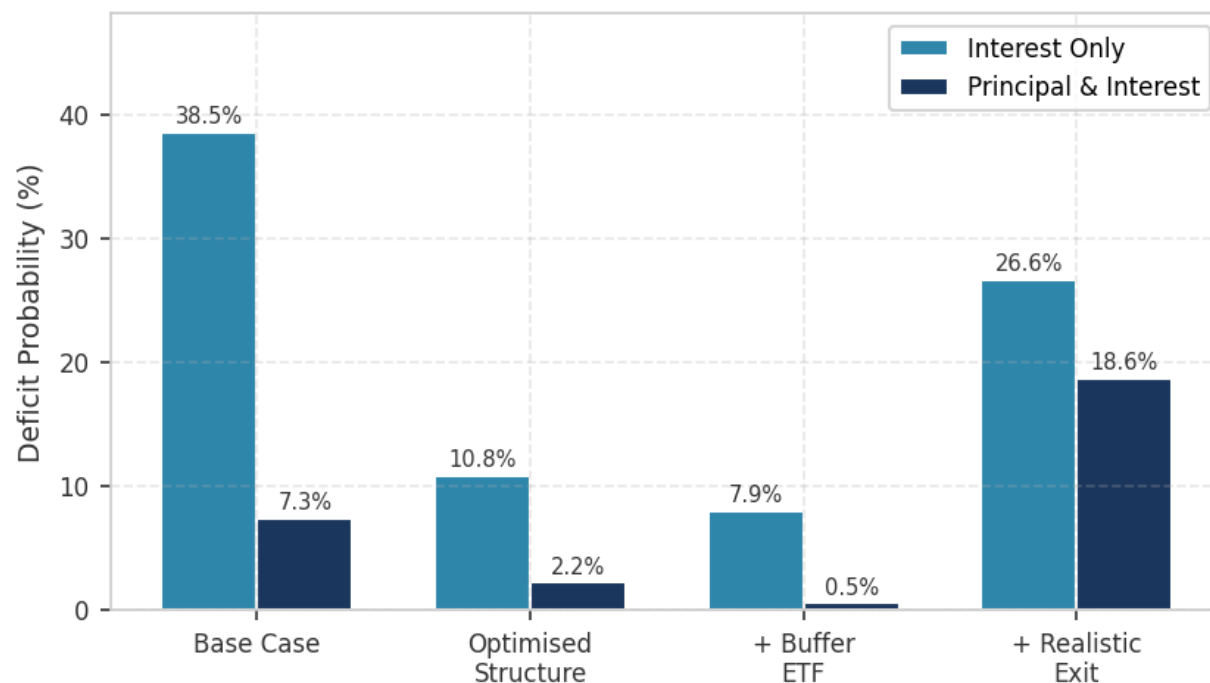
| Feature           | Detail                                          |
|-------------------|-------------------------------------------------|
| Loan Type         | Interest Only or Principal & Interest           |
| Typical Loan      | \$1.0-1.5M (60-80% LVR)                         |
| Investment        | BlackRock-managed equity portfolio (Buffer ETF) |
| Expected Return   | 10% p.a. (BlackRock assumption)                 |
| Annuity Income    | ~\$28,000/yr for 8 years                        |
| Profit Sharing    | 20% of gains every 5 years                      |
| Holiday Mechanism | Interest deferred during drawdowns              |
| Tenure            | Up to 30 years                                  |
| Insurance         | Built-in deficit protection                     |

### The Money Flow

1. Homeowner takes EPM loan (\$1.25M on \$2M property)
2. Loan funds invested in BlackRock Buffer ETF
3. Each quarter: investment returns pay interest + costs
4. Years 1-8: homeowner receives annuity income (\$28K/yr)
5. Every 5 years: profit share distributed to homeowner
6. At exit: surplus returned, loan repaid from investment balance

## Quantitative Validation — The Numbers

### Deficit Probability — All Configurations: IO vs P&I



### Independent Monte Carlo Results (50,000 paths)

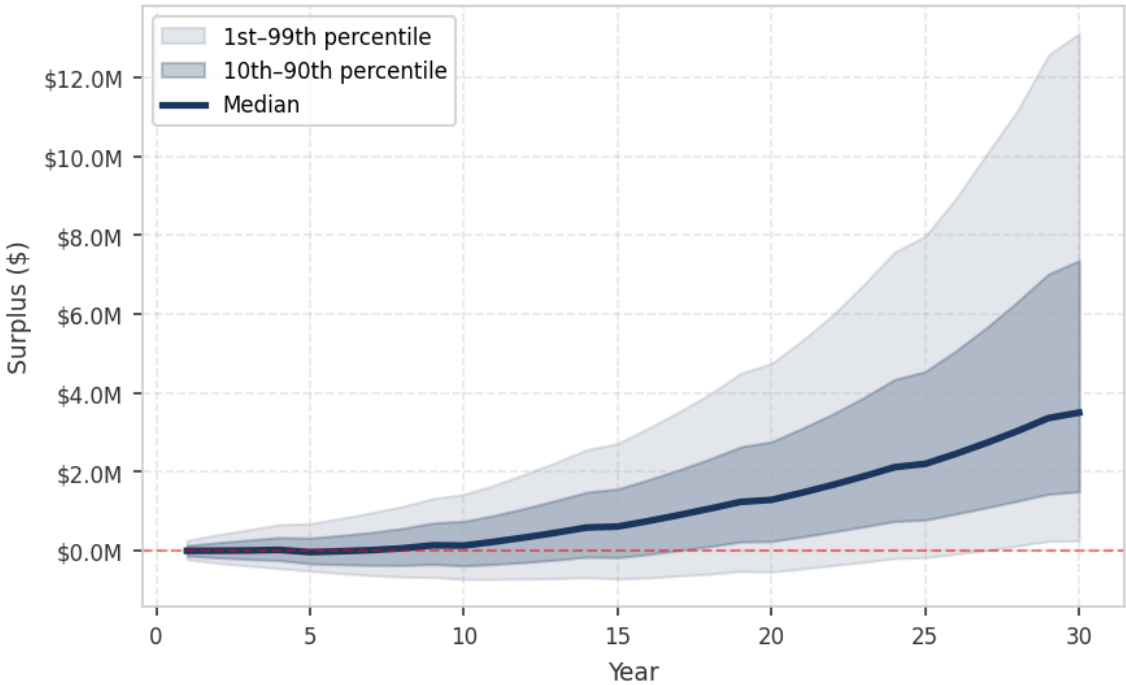
| Configuration       | IO Deficit | P&I Deficit | IO Premium | P&I Premium |
|---------------------|------------|-------------|------------|-------------|
| Base Case           | 38.5%      | 7.3%        | \$78,831   | \$10,056    |
| Optimised Structure | 10.8%      | 2.2%        | \$14,980   | \$2,794     |
| + Buffer ETF        | 7.9%       | 0.5%        | \$7,883    | \$426       |
| + Realistic Exit    | 26.6%      | 18.6%       | \$26,553   | \$15,116    |

### What This Means

- With P&I; + Buffer ETF + structural optimisation: only 0.5% of mortgages result in a deficit
- The fair insurance premium is just \$426 per EPM — negligible cost
- Mean surplus at maturity: \$4,083,007 per EPM (money returned to homeowner)
- The product generates positive outcomes for homeowners in 99.5% of scenarios

# Homeowner Wealth Creation

## Homeowner Wealth Creation — P&I + Buffer ETF (50,000 simulations)



The fan chart above shows the distribution of homeowner surplus over 30 years under the recommended P&I + Buffer ETF configuration. The median surplus grows steadily, reaching over \$3M by year 25. The 90th and 99th percentiles show substantial wealth creation potential — the product delivers significant value to homeowners in the vast majority of scenarios. The narrow band of the 10th-90th percentile range demonstrates the risk-compressing effect of the buffer ETF allocation.

## Revenue Model

### Revenue Streams per EPM (\$1.25M loan)

| Revenue Source                  | Estimated Annual | Over 20yr Life    |
|---------------------------------|------------------|-------------------|
| FutureProof margin (0.25%)      | \$3,125/yr       | \$62,500          |
| Origination fee (1.0%, upfront) | \$12,500         | \$12,500          |
| Insurance premium (upfront)     | \$426-\$7,883    | \$426-\$7,883     |
| Profit share participation      | Variable         | Variable          |
| Total per EPM                   |                  | \$75,000-\$83,000 |

### Portfolio Economics

| Scale   | EPMs  | Annual Revenue | Cumulative AUM |
|---------|-------|----------------|----------------|
| Year 1  | 50    | \$200K         | \$62M          |
| Year 3  | 500   | \$1.9M         | \$625M         |
| Year 5  | 2,000 | \$7.5M         | \$2.5B         |
| Year 10 | 5,000 | \$18.8M        | \$6.25B        |

*Note: Revenue grows with AUM as margin income compounds. Origination fees provide upfront cash flow. Insurance premiums are largely retained (80%) with 20% ceded to reinsurance.*

# Competitive Advantages & Moat

## 1. Product Innovation

- Only equity preservation mortgage in the Australian market
- Patent-pending holiday mechanism and equity structure
- Customers retain home ownership AND participate in investment upside
- Fundamentally different value proposition from traditional reverse mortgages

## 2. Institutional Partnerships

- BlackRock fund management — removes investment performance risk
- Buffer ETF allocation — institutional-grade risk management at no additional cost
- Reinsurance partnerships — transfers tail risk to capital markets

## 3. Quantitative Edge

- Proprietary 50,000-path Monte Carlo risk engine (Rust, 600K+ paths/sec)
- Ability to price insurance premiums with <1% standard error
- Real-time portfolio monitoring and stress testing capability
- Independent model validation demonstrates product viability

## 4. Regulatory & Market Position

- First-mover advantage in Australian equity preservation market
- Addresses a genuine unmet need (asset-rich, income-poor retirees)
- Positive social impact narrative for ESG-focused investors
- Growing demographic tailwind (ageing population, rising property values)



# Risk Factors

## Product Risks

- Equity market returns below 10% assumption → mitigated by buffer ETF and P&I structure
- Higher-than-expected wholesale funding costs → P&I viable even at 3% margin
- Regulatory changes to reverse mortgage / equity release rules
- Reputational risk if early EPMs underperform

## Execution Risks

- Scaling from concept to portfolio requires significant capital
- Wholesale funding negotiations ongoing (current quotes at 3%, target 1.5-2%)
- Insurance/reinsurance market appetite untested at scale
- Technology platform development required

## Market Risks

- Australian property market correction → affects LVR and collateral values
- Interest rate environment → model shows negligible sensitivity (validated)
- Competition from traditional reverse mortgage providers adapting their products
- Consumer education and trust-building required

## Mitigants

- P&I structure provides fundamental risk reduction (deficit 0.5% vs 38.5%)
- Buffer ETF compresses volatility while maintaining expected returns
- Diversified portfolio across cohorts and vintage years reduces concentration
- 50,000-path Monte Carlo engine enables precise pricing and risk management

## Summary & Next Steps

### Investment Thesis

FutureProof has developed a genuinely innovative financial product with validated economics, institutional partnerships, and a large addressable market. The quantitative analysis confirms the product works — particularly with P&I repayment and buffer ETF allocation, achieving a 99.5% success rate.

### Use of Funds

- Technology platform development (webapp, risk engine, portfolio management)
- Regulatory and legal framework
- Wholesale funding partnerships
- Initial EPM originations (pilot portfolio)
- Team expansion (sales, operations, compliance)

### Key Milestones

1. Finalise wholesale funding arrangement (target: 1.5-2.0% margin)
2. Launch pilot program (50 EPMs)
3. Establish reinsurance partnership
4. Scale to 500 EPMs within 3 years
5. Portfolio AUM target: \$625M by Year 3